WORKED SOLUTION

STEP I 2001, Question 10

Mechanics • Projectile Time of Flight

STEP I 2001, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q10

The key idea

Split the choice into two parts: the elevation fixes both the time of flight and the horizontal range, and the direction only decides *where* on a circle round the gun the shell comes down. So the shell can hit the line exactly when its range is at least d. The time of flight grows with the elevation but, beyond $45°$, the range shrinks, so we push the elevation up until the range has fallen to exactly d.

Solution

Time of flight and range for a given elevation

Model the shell as a particle moving under gravity alone (no air resistance), fired from ground level and landing on the plain. Fire it at elevation α, where $0 < \alpha \leq \frac{\pi}{2}$. Its vertical velocity component is $v\sin\alpha$ and its horizontal component is $v\cos\alpha$.

Vertically, its height after time t is $vt\sin\alpha - \frac{1}{2}gt^2$, which is zero again when

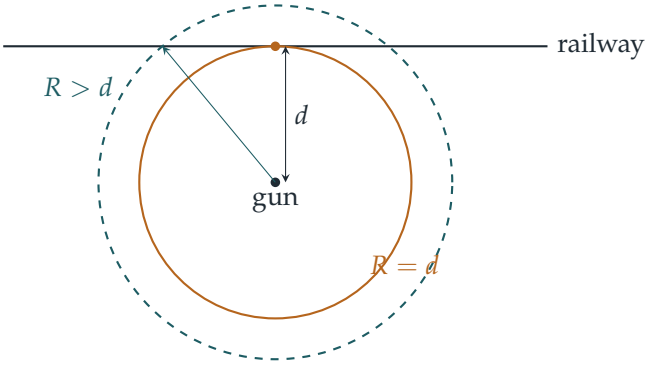
$$T = \frac{2v\sin\alpha}{g}.$$

In that time it travels a horizontal distance

$$R = v\cos\alpha \cdot \frac{2v\sin\alpha}{g} = \frac{v^2\sin 2\alpha}{g}.$$

When can the shell hit the line?

For a fixed elevation, turning the gun moves the landing point round a circle of radius R centred on the gun. The nearest point of the line is at distance d, and every other point of the line is further away. So the circle meets the line if and only if $R \geq d$. When $R = d$ the circle just touches the line, and the gun must point straight at the nearest point, perpendicular to the line.



(Seen from above. The dashed circle is a lower elevation, which reaches the line at two points. The solid circle is the optimal elevation, which just reaches it.)

So the gunner can hit the line at elevation α exactly when

$$\sin 2\alpha \geq \frac{gd}{v^2}.$$

Write $k = gd/v^2$. We're told $v^2 > gd$, so $0 < k < 1$ and some elevations do work. (The greatest range is v^2/g, at 45°, so this condition just says that the line is within range.)

Maximising the time of flight

Since $T = 2v \sin\alpha / g$ and $\sin\alpha$ is increasing on $(0, \frac{\pi}{2}]$, maximising T means choosing the *largest* allowed α.

For $0 < 2\alpha \leq \pi$, the condition $\sin 2\alpha \geq k$ holds exactly when

$$\arcsin k \leq 2\alpha \leq \pi - \arcsin k.$$

So the largest allowed elevation is $\alpha = \frac{\pi}{2} - \frac{1}{2}\arcsin k$, which is more than 45°. At this elevation $\sin 2\alpha = k$, and since 2α lies in $(\frac{\pi}{2}, \pi)$, $\cos 2\alpha$ is negative:

$$\cos 2\alpha = -\sqrt{1 - k^2}.$$

Ask yourself Wouldn't it be better to aim along the line at some distant point, so the shell flies further?

No. Flying further needs a *larger* range, and above 45° that means a *lower* elevation and so a shorter flight. Time of flight depends only on elevation, so the best we can do is the steepest shot that still reaches the line. Its range is exactly d, so the shell lands at the foot of the perpendicular from the gun to the line.

Now we square T and use $2\sin^2\alpha = 1 - \cos 2\alpha$:

$$\begin{aligned} g^2T^2 &= 4v^2\sin^2\alpha \\ &= 2v^2\left(1-\cos 2\alpha\right) \\ &= 2v^2 + 2v^2\sqrt{1-\frac{g^2d^2}{v^4}} \\ &= 2v^2 + 2\left(v^4 - g^2d^2\right)^{1/2}, \end{aligned}$$

taking v^2 inside the square root (allowed because $v^2 > 0$). This is the required result.

Checks

- As $d \to 0$ the formula gives $g^2T^2 \to 4v^2$, so $T \to 2v/g$. That's firing straight up, the longest possible flight, which is right when the line passes right by the gun.
- When $v^2 = gd$ the line is only just in range: the only possible shot is at 45°, and the formula gives $g^2T^2 = 2v^2$, which matches $T = 2v\sin 45^\circ/g = \sqrt{2}\,v/g$.
- Numerically, with $g = 9.8\,\mathrm{m\,s^{-2}}$, $d = 100\,\mathrm{m}$ and $v = 40\,\mathrm{m\,s^{-1}}$, the best elevation is about 71.1° and the formula gives $T \approx 7.72\,\mathrm{s}$. Stepping the motion numerically gives the same time, with the shell landing on the line. A brute-force search over all elevations agrees with the formula in hundreds of random cases.

Common slips

- Taking the lower root, $\alpha = \frac{1}{2}\arcsin k$. That shot also hits the line, but it gives the *shortest* time of flight, not the longest: in effect, taking the wrong sign for $\cos 2\alpha$.
- Assuming the shell must land at the nearest point of the line from the start. That turns out to be true, but it needs the argument above: any other point needs a larger range, and so a lower elevation.

Beyond STEP

This is a small constrained optimisation: maximise one quantity (flight time) subject to a constraint (reaching the line), and the optimum lies on the boundary of what's allowed. In first-year mechanics and optimisation courses you'll meet the general tools for this, such as Lagrange multipliers, and the related idea of the *envelope* of all trajectories (the "parabola of safety"), which bounds every point the gun can reach. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen your projectile and particle mechanics.

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